

Abdullah Al-Nafisah

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MSc, Electrical and Computer Engineering

2022 – 2024

King Abdullah University of Science and Technology (KAUST), Thuwal, KSA | GPA 3.83/4.0

Thesis: Turbo-Equalized FTN-Inspired Modulations, doi:10.25781/KAUST-M5140

BSc, Double Major in Electrical Engineering & Physics

2013 – 2019

King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, KSA | GPA 3.76/4.0, 1st Honors

Exchange Student

Fall 2016

Georgia Institute of Technology, Atlanta, GA, USA | GPA 3.5/4.0

Summer School on Superconducting Electronics

Summer 2024

FLUXONICS, The European Foundry for Superconducting Electronics e.V., Corsica, France

Research & Development

Oct 2020 – Present

National Company of Telecommunications and Information Security (NTIS), KSA

- **Qubit control on the RFSoc:** General-purpose RF system on the AMD/Xilinx RFSoc ZCU216 with Pulse and CW modes and native support for LOs, VNAs, digitizers, and AWGs; output power and spurious emissions characterized.
- **Real-time classification in fabric:** FPGA-based flexible line qubit classifier enabling mid-circuit measurement and active reset at nanosecond latency, with a trigger mode driven from a TTL-compatible sync signal to mitigate vibration-induced errors.
- **Qubit characterization:** Single-qubit calibration and characterization (resonator punch-out, two-tone spectroscopy, Rabi, Ramsey) with a trained single-shot readout discriminator; resonators measured at high and low power to estimate single-photon drive power.
- **Quantum links:** FPGA firmware and high-speed acquisition for a prototype QKD transmitter and receiver, and FPGA entropy extraction for QRNG on the IDQ20MC1 chip, certified to NIST SP 800-90B.
- **RF and embedded:** Direct RF sampling with x16 DACs and ADCs on RFSoc, optical links on MPSoC, RF switches built from off-the-shelf parts for RF-multiplexing, PCB design, and VHDL/Verilog controllers with custom UART, I2C, and SPI drivers.

Avionics Engineer

Dec 2019 – Oct 2020

PSATRI, Riyadh, KSA

- PCB design for unmanned aerial vehicle avionics in Altium, with drone weight, thrust-to-weight ratio, and battery sizing optimized for flight time.

Master's Thesis Research

Spring 2023 – Spring 2024

Supervised by Dr. T. Al-Naffouri and Dr. M. Siala at KAUST, Thuwal, KSA

- Integrated FTN-inspired modulations with turbo equalization and LDPC decoding, beating the BER reported in state-of-the-art literature.

Guided Research on Analog Electronic Circuits

Fall 2018 – Spring 2019

Supervised by Dr. M. Abuelma'atti at KFUPM, Dhahran, KSA

- Built a memristor-based chaotic masking system from off-the-shelf components, validated synchronization of two chaotic oscillators, and designed a tunable CFOA comb filter.

Mitacs Globalink Research Internship

Summer 2018

Supervised by Dr. S. Valluri at Western University, London ON, Canada

- Investigated the Lambert W function for the I-V characteristics of diode-based circuit models.

Visiting Student Research Program (VSRP) Internship

Summer 2017

Supervised by Dr. H. Fariborzi at KAUST, Thuwal, KSA

- Modeled All Spin Logic circuits in Cadence with Verilog-A, applying the Landau-Lifshitz-Gilbert equations and compact charge/spin conduction modules.

Projects

github.com/AbdullahAlNafisah

- **llm-kv260-demo:** FPGA-based edge AI, an LLM decoded by four AMD Kria KV260 FPGAs on one stream.
- **coded-msprs:** Rate-2 turbo-equalized MS-PRS simulation chain in C++ on AFF3CT, behind the thesis and the NSM paper.

Papers

1. M. Siala, A. Al-Nafisah, and T. Al-Naffouri, "Nyquist Signaling Modulation (NSM): An FTN-Inspired Paradigm Shift in Modulation Design for 6G and Beyond," *arXiv:2511.08553*, 2025.
2. A. Al-Nafisah, D. Khan, S. Amara, and Y. Massoud, "Synchronization of Two Transistor-Based Chaotic Oscillators: Experimental Verification," *IEEE 65th International Midwest Symposium on Circuits and Systems (MWSCAS)*, Fukuoka, Japan, pp. 1–4, 2022.
3. M. T. Abuelma'atti and A. Al-Nafisah, "New CFOA-Based Shadow Analog Comb Filter," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 1–8, 2020.
4. M. T. Abuelma'atti and A. Al-Nafisah, "Synchronization of Memristor-Based Chaotic Oscillator: Experimental Verification," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 21–28, 2020.
5. M. T. Abuelma'atti and A. Al-Nafisah, "A Memristor-Based Chaotic-Masking for Analog Spread-Spectrum Communication," *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 14, no. 2, pp. 966–971, 2019.